Economics of Health and Health Care

Lectures 4 and 5 - Demand for health and health care

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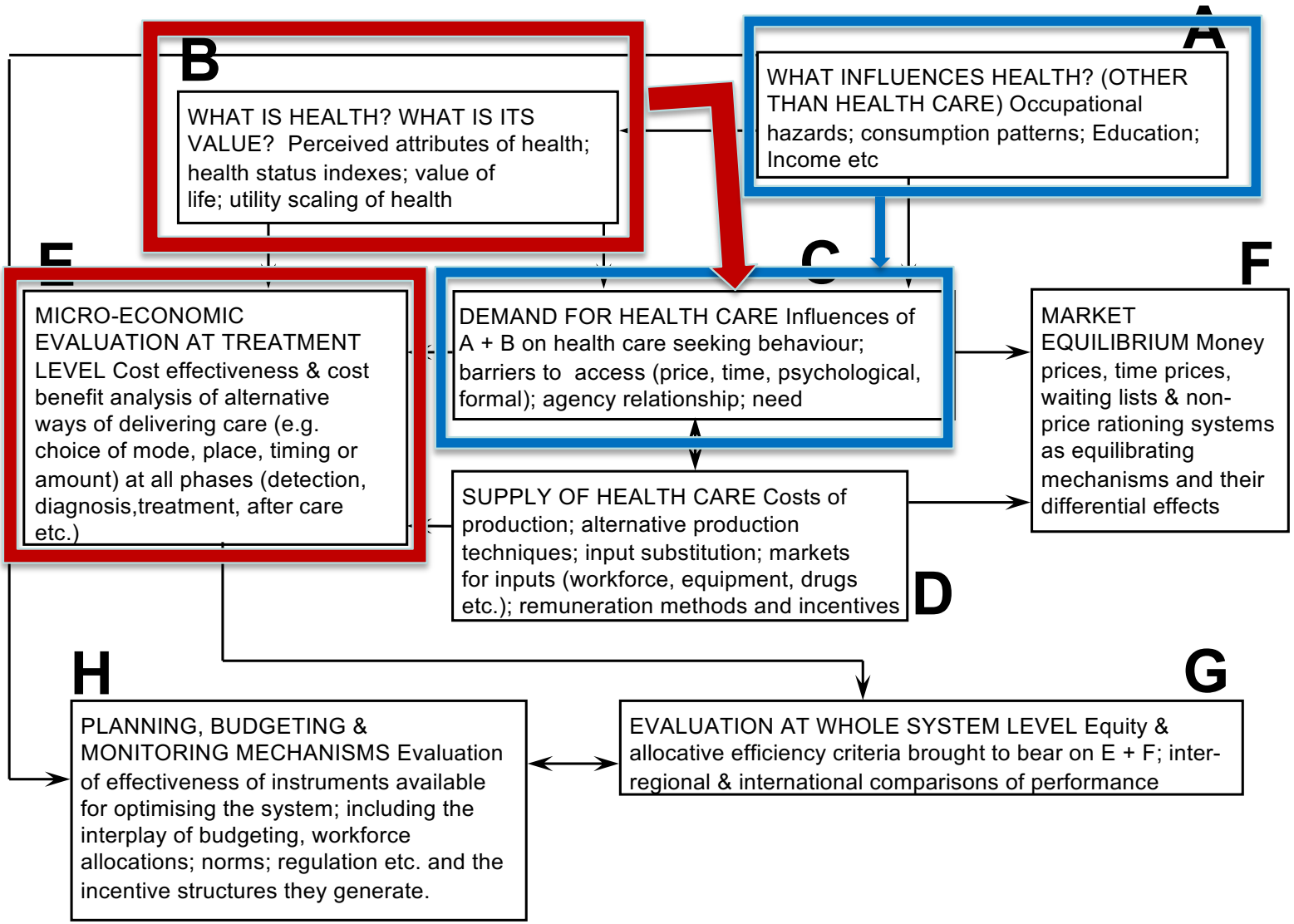
B
WHAT IS HEALTH? WHAT IS ITS VALUE? Perceived attributes of health; health status indexes; value of life; utility scaling of health
A
WHAT INFLUENCES HEALTH? (OTHER THAN HEALTH CARE) Occupational hazards; consumption patterns; Education; Income etc
E
MICRO-ECONOMIC EVALUATION AT TREATMENT LEVEL Cost effectiveness & cost benefit analysis of alternative ways of delivering care (e.g. choice of mode, place, timing or amount) at all phases (detection, diagnosis,treatment, after care etc.)
C
DEMAND FOR HEALTH CARE Influences of A + B on health care seeking behaviour; barriers to access (price, time, psychological, formal); agency relationship; need
F
MARKET EQUILIBRIUM Money prices, time prices, waiting lists & non-price rationing systems as equilibrating mechanisms and their differential effects
SUPPLY OF HEALTH CARE Costs of production; alternative production techniques; input substitution; markets for inputs (workforce, equipment, drugs etc.); remuneration methods and incentives
D
H
PLANNING, BUDGETING & MONITORING MECHANISMS Evaluation of effectiveness of instruments available for optimising the system; including the interplay of budgeting, workforce allocations; norms; regulation etc. and the incentive structures they generate.
G
EVALUATION AT WHOLE SYSTEM LEVEL Equity & allocative efficiency criteria brought to bear on E + F; inter-regional & international comparisons of performance

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Outline

- Demand for health and health care
  - Grossman model
  - Comparative statics in the Grossman model
  - Empirical evidence
- Demand for health care – partial equilibrium analysis
  - Role of time
  - Role of insurance / the Rand Health Insurance experiment
  - Income effects

Micro analysis

- We have seen what is the value of health
- Time to address individual behavior
- Describe the demand for health and health care as the result of individual choices

Grossman model

The main microeconomic model of health choice and demand for health and for health care

What's new:

- See health as a stock
- See health as a joint production good, which requires both the contribution of the individual (time) and consumption of other goods and services - health care

Implications of the Grossman model

- Health care is an intermediate good
- If health is a stock, it lasts for more than one period - there is a depreciation rate (which may be stochastic)
- Health must be seen as an investment good AND as a consumption good

The Grossman model

- It is originally defined as an intertemporal maximization problem - one must consider the full lifetime horizon
- Take a simple version of the model, without the time dimension
- This allows to explore similarities with other models and prepare your economic intuition

The model - key elements

- ***U=U(B, H)*** - utility function (this version does not consider the disutility of working and the cost of sickness days)

- ***H=H(M,T_H; E, A)*** - Health production - what resources are used, what influences the ability to produce health

- ***B=B(X, T_B; E)*** - consumption function (can be simplified to consumption), **B= X**

The model - key elements: constraints

time available to

- Work (and earn a wage, providing income)
- Produce health
- Produce consumption (we will ignore this one)
- Spend as sickness days

- Two conditions:
  - Respect total time
  - Respect income available

The Grossman model

First analogy: choice between work and leisure, for a given wage

What is different here?

- The number of days available for work is an endogenous variable, determined by the stock of health
- Other than that, we have the same trade-offs
- Health and consumption decisions will be related to each other.

The Grossman model

Consumer's problem:

$$\max_{\{T_H, T_B, T_W, T_L, X, M\}} U(B,H)$$

$$s.t. \quad B = B(X, T_B; E)$$

$$H = H(M, T_H; E, A)$$

$$T_B + T_H + T_W + T_L = T*$$

$$wT_W = pX + M$$

$$T_L = f(H), f' < 0$$

- Utility from consumption and health
- Consumption definition
- Health production function
- Time constraint
- Budget constraint
- sickness days

The Grossman model

The constraints define a possibility set for health and consumption outputs (H and B)

We can simplify it, using the constraints to substitute for B and H in the objective function, and to substitute $f(H)$ and the monetary budget constraint.

The endogenous variables are then: T_H, T_B, X, M

Implications of the model

- **[age]** in this model, death is determined endogenously; depends on the depreciation rate of the health stock and how this changes with age
- It is consistent with the elderly being more intensive users of health care - the health stock depreciates faster, they invest more than younger people but the first effect is still dominant
- (note that health and demand for health care evolve differently with age – this can be tested with data)

Implications of the model

- **[wage]** an increase in the wage leads to an increase in the health stock
- For the same consumption of intermediate goods prior to the wage increase, the consumer has now more free time – to earn the same money, needs to work less hours.
- Consequently, (s)he can invest more in both production of health and consumption – more health, more demand for health care (everything else constant)

Implications of the model

- One should expect that consumers with higher wages have a higher health stock, which for equal depreciation rate means a higher demand of health care
- This is a testable implication - an income effect
- If the poor have a higher depreciation rate due to other factors (say, poor nutrition), they may consume more health care - thus we need to be careful in the analysis
- income/wage effects on health and on health care can be different

Implications of the model

- **[education]** can be seen as something that increases the efficiency of health production given the health investment and the increases the efficiency of producing the consumption good
- The consumer with a higher level of education will select to have a higher health stock, for the same “inputs”
- To achieve the same health as someone less educated, needs less health care, given the higher efficiency in “health production”
- The consumer may also demand more health care, as the marginal efficiency of the intermediate input is higher

Implications of the model

Demand for health care is a *derived demand*

Health benefits consumers due to

- Higher utility if in good health
- Less sickness days, allowing for more time to allocate to work, leisure and production of health and consumption
- Higher productivity per unit of time
- Higher life expectancy

Implications of the model

- Does the theory make sense?
- People do not look for two apendicitis or cardio-toraccic intervention as a response to a lower price in surgery
- Once one recognizes that often the demand for health care corresponds to unique episodes, does it make sense to believe prices to have a role?
- There is intensive vs extensive margin - more people doing it versus the same person having more of it

Empirical work

- A crucial element is the health production function
- Can we observe it?
- Estimates of a production function - simple example due to Kenkel, 1995, Health Economics
- Replicate it for Portuguese date

Production function of health

- Kenkel uses an econometric model to assess the importance of life-style variables to the production of health
- The work of Kenkel focus on
  - Taking breakfast
  - Smoking
  - Doing exercise

Starting point: factors that favour a better health status (study Alameda 7)

- Keep “right” weight
- Do not eat between meals
- Do not smoke
- Do exercise on a regular basis
- Do not drink (or do it moderately)
- Do take breakfast
- Sleep 7-8 hours regularly

Methodology

- Variables - characteristics of individuals, life-style choices
- Health status is captured by a latent variable (not directly observed)

  $H^* = g(factors)$

- We observe a partition of the health status variable
- There are thresholds from one observed health status to another (with 5 health status, we have four thresholds)

Econometric implications

- Without loss of generality, one of the limits can be set to zero
- Parameters can be estimated up to a scale factor. Normalization of standard error to 1
- It results from the fact that the latent index does not have a unit of measure

Main results

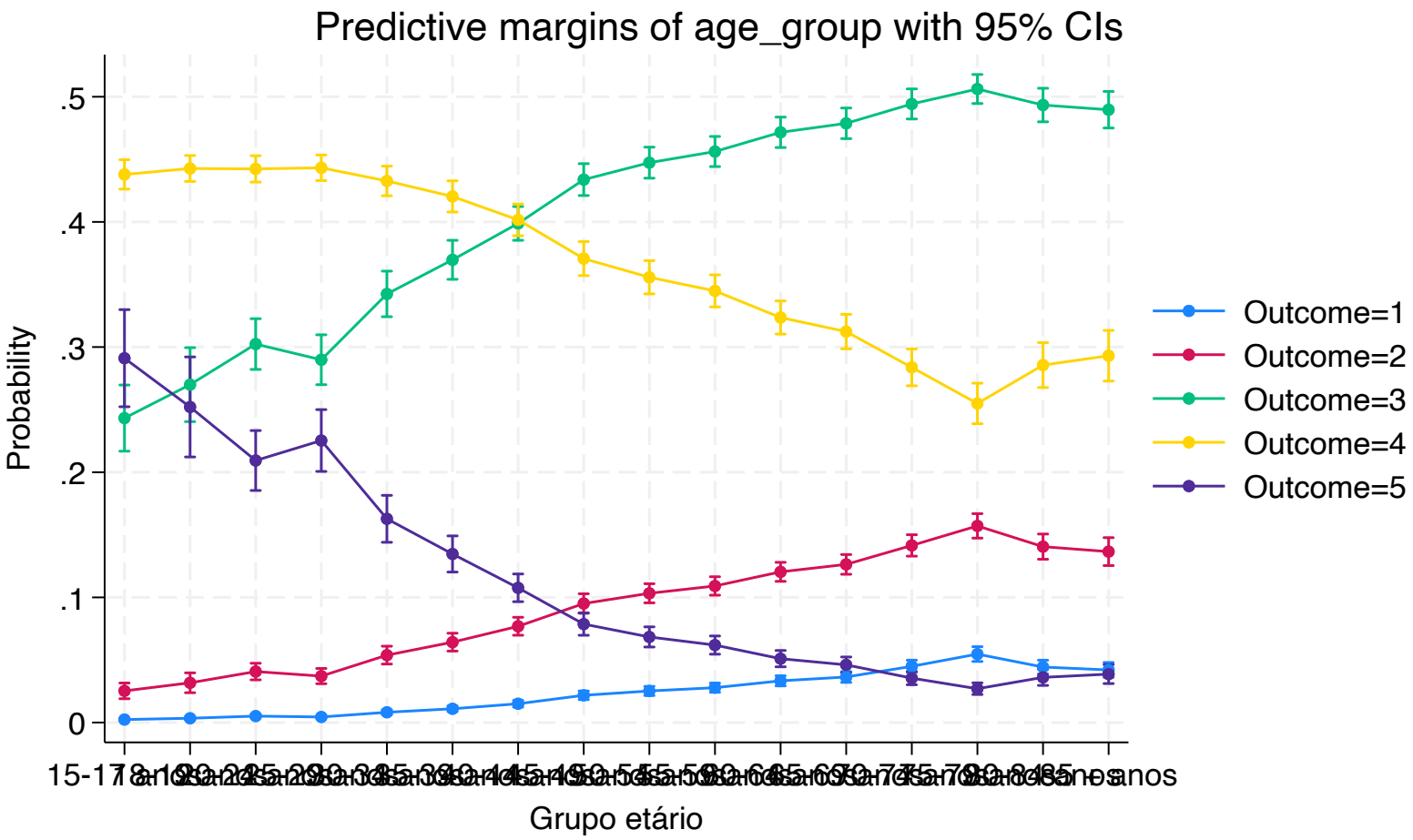
- Weight (-, after some level)
- Smoking (-)
- Drinker (-)
- Stress (-)
- Heart problems (-)
- Diabetes (-)
- Age (-)
- Schooling (+)

Ordered probit regression

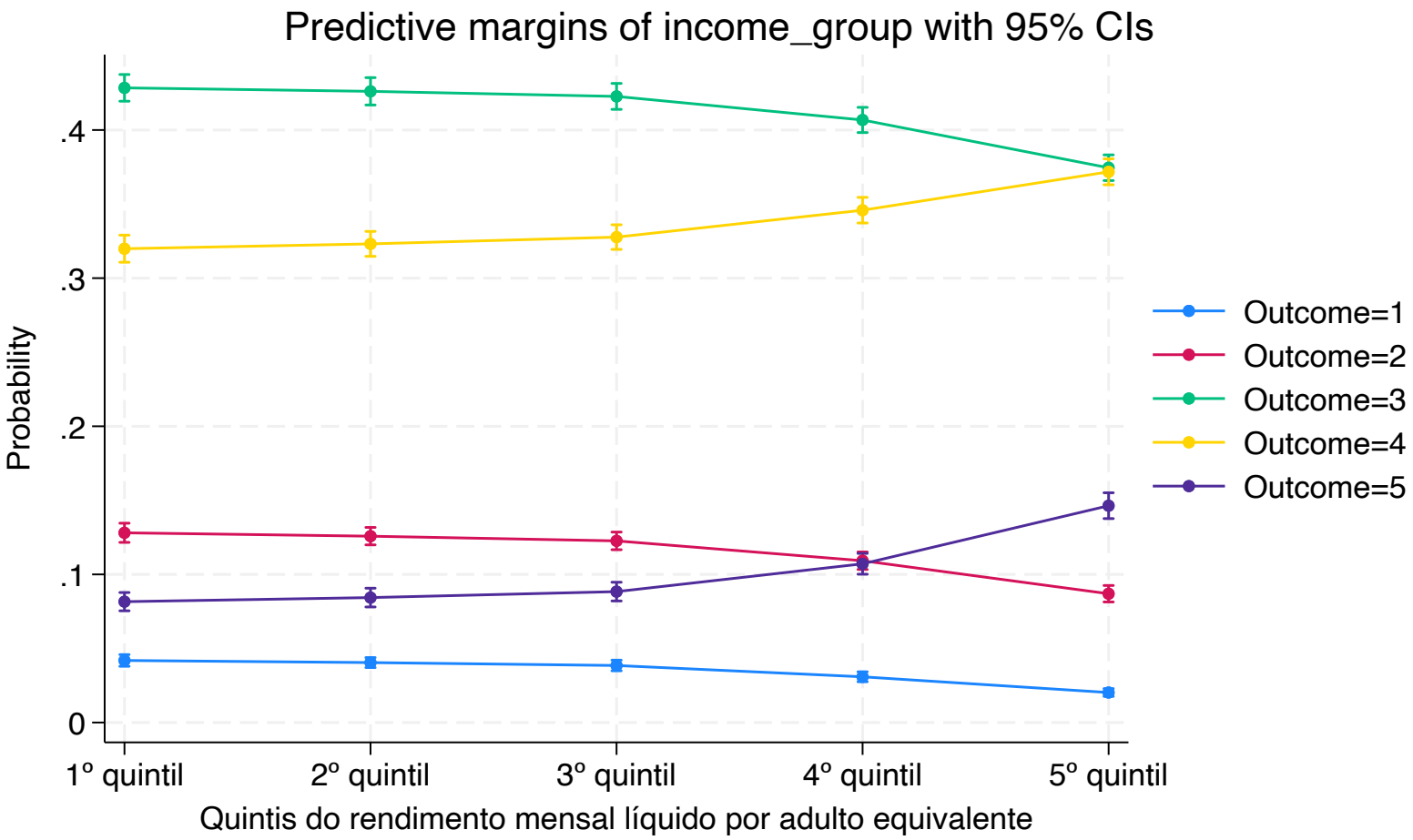
Number of obs =	13,918
LR chi2(42) =	10712.07
Prob > chi2 =	0.0000
Pseudo R2 =	0.2861

Log likelihood = -13366.398

health	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
age_group						
18-19 anos	-.1533241	.1073979	-1.43	0.153	-.3638201	.0571719
20-24 anos	-.3341666	.0891923	-3.75	0.000	-.5089802	-.1593529
25-29 anos	-.264835	.091986	-2.88	0.004	-.4451241	-.0845458
30-34 anos	-.5544235	.0887188	-6.25	0.000	-.7283091	-.3805379
35-39 anos	-.7053687	.0845176	-8.35	0.000	-.8710202	-.5397172
40-44 anos	-.8709078	.0823088	-10.58	0.000	-1.03223	-.7095855
45-49 anos	-1.083052	.0825156	-13.13	0.000	-1.244779	-.9213244
50-54 anos	-1.171677	.083317	-14.06	0.000	-1.334975	-1.008379
55-59 anos	-1.233048	.0838118	-14.71	0.000	-1.397316	-1.06878
60-64 anos	-1.347038	.0848533	-15.87	0.000	-1.513347	-1.180728
65-69 anos	-1.405154	.085616	-16.41	0.000	-1.572958	-1.23735
70-74 anos	-1.548269	.0865501	-17.89	0.000	-1.717904	-1.378634
75-79 anos	-1.689773	.0879965	-19.20	0.000	-1.862243	-1.517304
80-84 anos	-1.539351	.0901164	-17.08	0.000	-1.715975	-1.362726
85 + anos	-1.50229	.093123	-16.13	0.000	-1.684808	-1.319772
gender						
Mulher	-.0677636	.022054	-3.07	0.002	-.1109887	-.0245386


Predictive margins of age_group with 95% CIs
Probability
.5
.4
.3
.2
.1
0
Grupo etário
Outcome=1
Outcome=2
Outcome=3
Outcome=4
Outcome=5

income_group						
2º quintil	.0243163	.0313572	0.78	0.438	-.0371427	.0857752
3º quintil	.0588637	.0330751	1.78	0.075	-.0059622	.1236896
4º quintil	.207052	.0344876	6.00	0.000	.1394575	.2746465
5º quintil	.4696818	.0372009	12.63	0.000	.3967693	.5425942
1.SNS	.4771238	.433217	1.10	0.271	-.371966	1.326214
1.subsistema_pu~o	.6504818	.43383	1.50	0.134	-.1998094	1.500773
1.subsistema_pr~o	.6543249	.4363081	1.50	0.134	-.2008233	1.509473
1.private_insur~e	.1588671	.0254907	6.23	0.000	.1089063	.2088279
1.doença_crónica	.7053084	.0238291	29.60	0.000	.6586042	.7520126
Daily_Life_Activ						
2	.959583	.0411037	23.35	0.000	.8790213	1.040145
3	1.99669	.0440681	45.31	0.000	1.910318	2.083062
underweight	-.2056862	.0772314	-2.66	0.008	-.357057	-.0543154
overweight	-.0453479	.0223814	-2.03	0.043	-.0892146	-.0014813
obesity	-.1261631	.0261453	-4.83	0.000	-.1774069	-.0749192


Predictive margins of income_group with 95% CIs
Probability
.4
.3
.2
.1
0
1º quintil
2º quintil
3º quintil
4º quintil
5º quintil
Quintis do rendimento mensal líquido por adulto equivalente
Outcome=1
Outcome=2
Outcome=3
Outcome=4
Outcome=5

daily_meals						
2	.1498908	.114553	1.31	0.191	-.074629	.3744105
3	.2277404	.1114002	2.04	0.041	.0094001	.4460808
tabaco						
Fuma ocasional..	.0180872	.0711121	0.25	0.799	-.12129	.1574644
Já fumou, mas ..	.0235857	.0338508	0.70	0.486	-.0427606	.089932
Não fuma	.0172441	.0306317	0.56	0.573	-.042793	.0772812
1.cope_unexpect~e	.2044475	.0222925	9.17	0.000	.1607551	.2481399
/cut1	-1.221566	.4524938			-2.108437	-.334694
/cut2	.0080918	.4524703			-.8787336	.8949173
/cut3	1.991689	.4529487			1.103926	2.879452
/cut4	3.646319	.45319			2.758083	4.534555

Partial equilibrium analysis

Grossman model - individual demand for health care

Implications for demand for health care in partial equilibrium

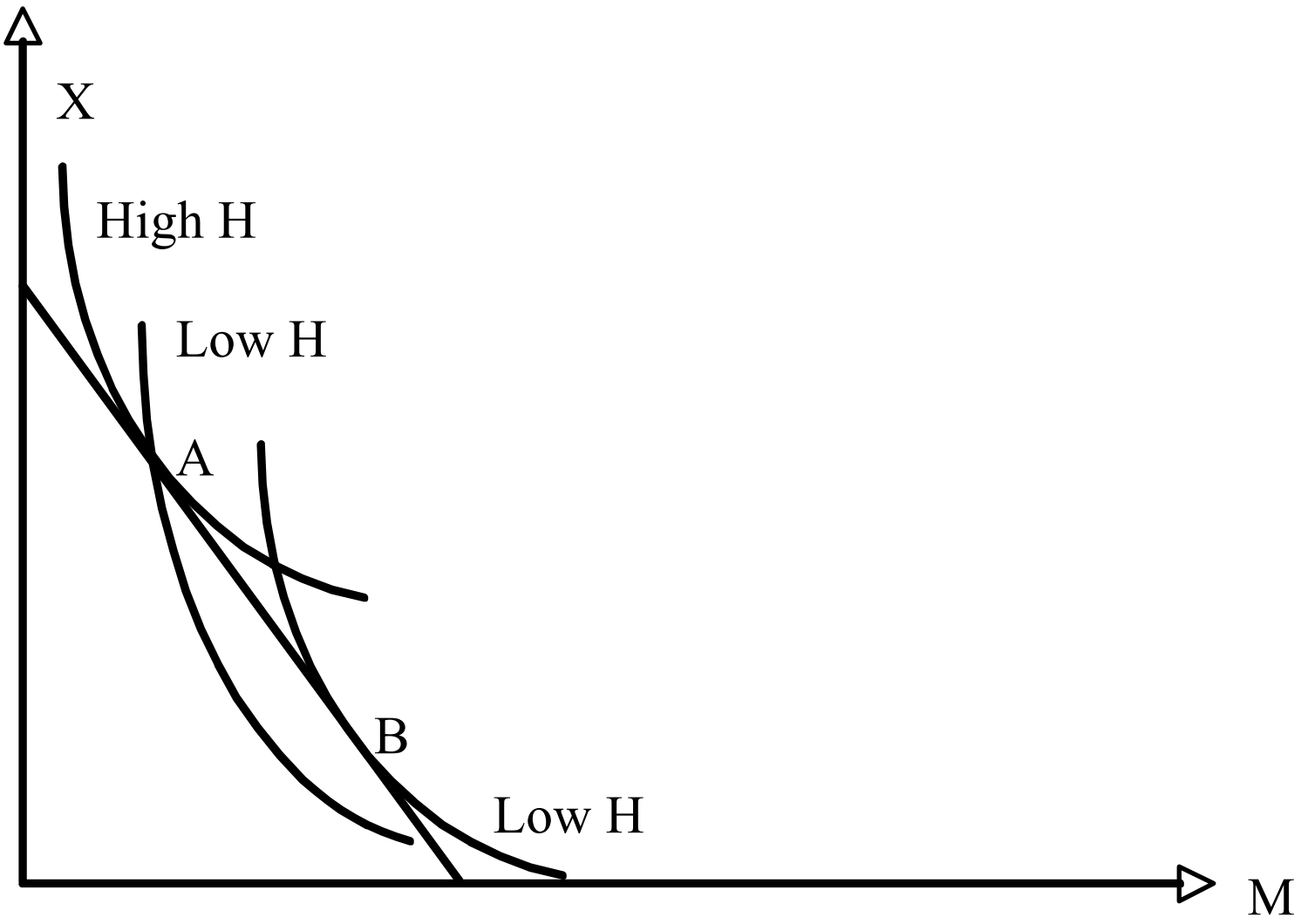
Demand curve - shifters

- Waiting time
- Insurance coverage
- income

The demand curve

Demand for health care - state-dependent utility:

- the willingness to sacrifice consumption of other goods may be different according to the health status
- Most demand for health care takes place under the state “being sick”
- Once state dependence is acknowledged usual analysis can be applied


X
High H
Low H
A
B
Low H
M

Role of waiting time

“prices” of health care must include time

Distinction: treatment time and access time

How important is it?

Early studies: Acton in 1970s, other have been done since

Time demand elasticity - early evidence

- Acton (1975, 1976)
  - Models the number of visits for various types of care as determined by travel time, waiting time, income and other socio-economic variables
  - Results:
    - -0,958 for outpatient visits (in the hospital)
    - -0,252 for private physician visits
- Coffey (1983)
  - Elasticity of time price with respect to the probability of seeking care is -0.1
  - Elasticity of time price with respect to the quantity of care is also -0.1
- Mueller and Monheit (1988)
  - Find inelastic time price elasticity with respect to the probability of seeking dental care but relatively elastic time price elasticity with respect to the quantity of dental care

More recent

- P. Sivey, The effect of waiting time and distance on hospital choice for English cataract patients, 2012, Health Economics
- Estimates the waiting time elasticity of demand
- (GPs and patients) choose public hospital (not a choice to forego treatment or to to private sector)
- approximately -0.1 (overall average), meaning 10% increase in waiting time, decrease in demand of 1%

More recent

- But there is considerable heterogeneity (two groups):

Table IV. Latent class multinomial logit model with practice rurality determining class probability

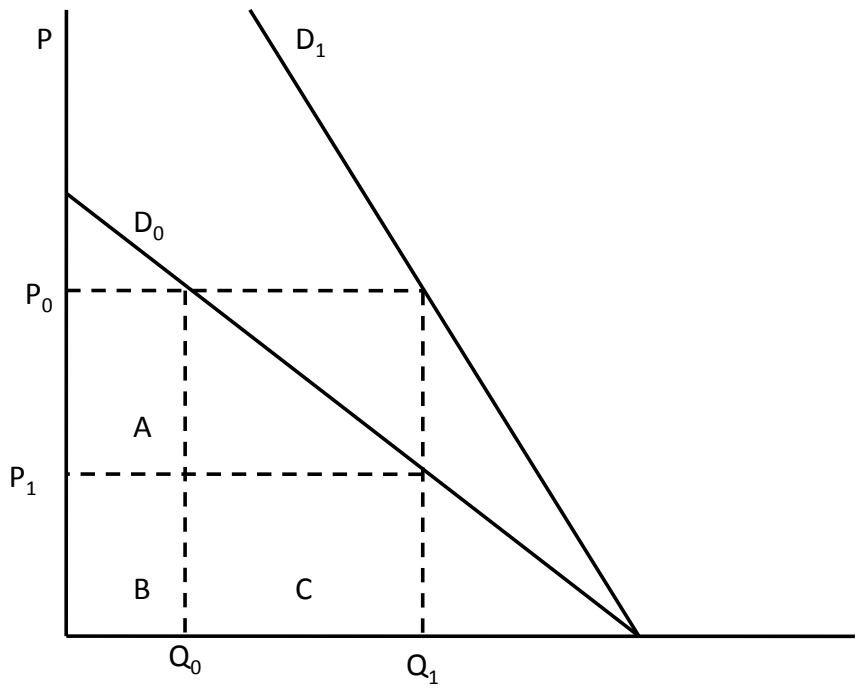
	Model 4			
	Class 1		Class 2	
Variable	Coeff.	S.E.	Coeff.	S.E.
Ln(travel time)	−8.673***	0.055	−3.074***	0.021
Inpatient wait	−0.016***	0.006	−0.109***	0.006
Outpatient wait	−0.125***	0.038	−0.269***	0.037
Class membership probability	0.691		0.309	
Constant			−0.804***	0.070
Rural practice			0.590***	0.174

Values more travel time (more sensitive to it) than waiting time

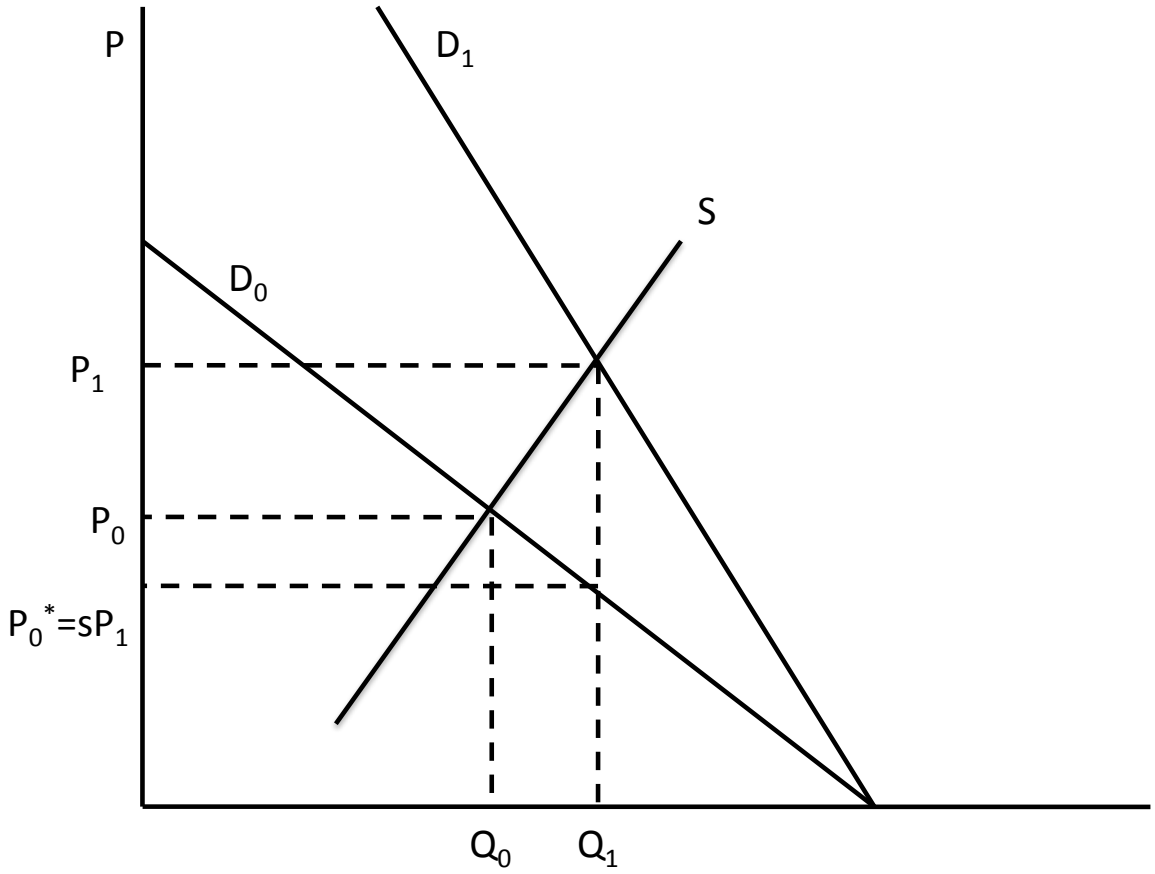
Values more waiting time than travel time

Another driver: insurance coverage

- Insurance coverage reduces the price to be paid by the consumer at the moment of health care utilization
- This, according to standard economic intuition should increase demand
- Let co-insurance rate be the fraction of price paid by the patient, let co-payment be the total value paid by the patient
- If P is the price, s is the co-insurance rate, then it follows that sP is the co-payment



- Insurance coverage creates a difference between the price paid by the consumer and the price received by the health care provider
- A reduction in the insurance coverage may increase or decreases co-payments, depending on the price elasticity (smaller or greater than 1)



- There is also a market Effect
- To increase supply, a higher price will emerge

RAND Health Insurance Experiment

- Controlled experiment on the effect of insurance on the consumption decisions, directly addresses price sensitivity
- Different health plans have been assigned randomly
- The differences across plans were on the copayment required
- It ranged from zero copayment to full payment (with an overall cap)

RAND Health Insurance Experiment

- Design of the experiment
- Approximately 2000 nonelderly families were assigned to insurance plans
- Families came from 6 areas of the country and participated for 3 or 5 years
- While they participated, the families used only the insurance plan administered by the Experiment
- Participation Incentive payment insures no family would become financially worse off

RAND health Insurance Experiment

- Health plans offered:
  - Free care
  - 25% coinsurance*
  - 50% coinsurance*
  - 95% coinsurance*
  - Deductible – free care for inpatient services and 95% coinsurance for outpatient services, up to a limit of 150 USD per person or 450 USD per family

*With a cap on family out-of-pocket expenditures depending on the family's income

RAND - health Insurance Experiment

- Dependent variable
  - Medical expenditure including all services except dental and outpatient mental health expenditure
  - But also:
    - Annual number of face-to-face visits
    - Probability of any use of medical services
    - Hospital admission rate

RAND Health Insurance Exp.

- Explanatory variables:
  - *The health insurance plan*
    - 5 categories for the estimation of total expenditure and 3 categories for the analysis of inpatient use
  - *The health status* (evaluated before the beginning of the program)
    - Measured by General Health Indexes, the existence of chronic illnesses, permanent disabilities and Mental Health Index
  - *Economic and socio-demographic variables*
    - Includes sex, age, race, family income and family size

RAND Health Insurance Exp.

- Unit of analysis
  - Person-year
- Statistical Methods
  - Analysis of variance (ANOVA)
  - Estimation of demand for medical services is based on a four-equation model
- Because ANOVA yields consistent but unprecise estimates due to extremely assymetric distribution of health care expenses (1/6 to ¼ of the sample had no medical expenses)

RAND Health Insurance Exp.

	Likelihood of any use (%)	Prob of any inpatient use (%)	Medical Expenses per person (1991 $)
0%	86,7	10,4	1019
25%	78,8	8,83	826
50%	74,3	8,31	764
95%	68,0	7,75	700
Deductible	72,6	9,52*	817

Adjusted annual emergency department use per person

	Expense (1991$)	Prob of any use	Visits per 1000 persons
0%	100%=$42	100%=0.22	100%=304
25%	86	85	79
50%	90*	92*	82*
95%	70	70	65
Deductible	82	81	80

*The reversal in order involving the 50% plan is not statistically significant

The deductible case had free emergency case so the results are unsurprising

RAND – Health Insurance Experiment

- Responsiveness to plan varied with the kind of medical problems for which patients use the emergency department (ED)
  - The use of ED involving more urgent diagnoses was 23% lower on the plans with any cost-sharing than on the free plan
  - But it was 47% lower on less urgent diagnoses

RAND Health Insurance Exp.

- The is an important impact of coinsurance rate on the level of medical care expenses:
  - As we move from 95% coinsurance to free care, expenses increase about 50%
- Coinsurance rates seems to help self-sorting in the use of the Emergency Department
- Prices and coinsurance rate as thus two important factors affecting the demand for medical care

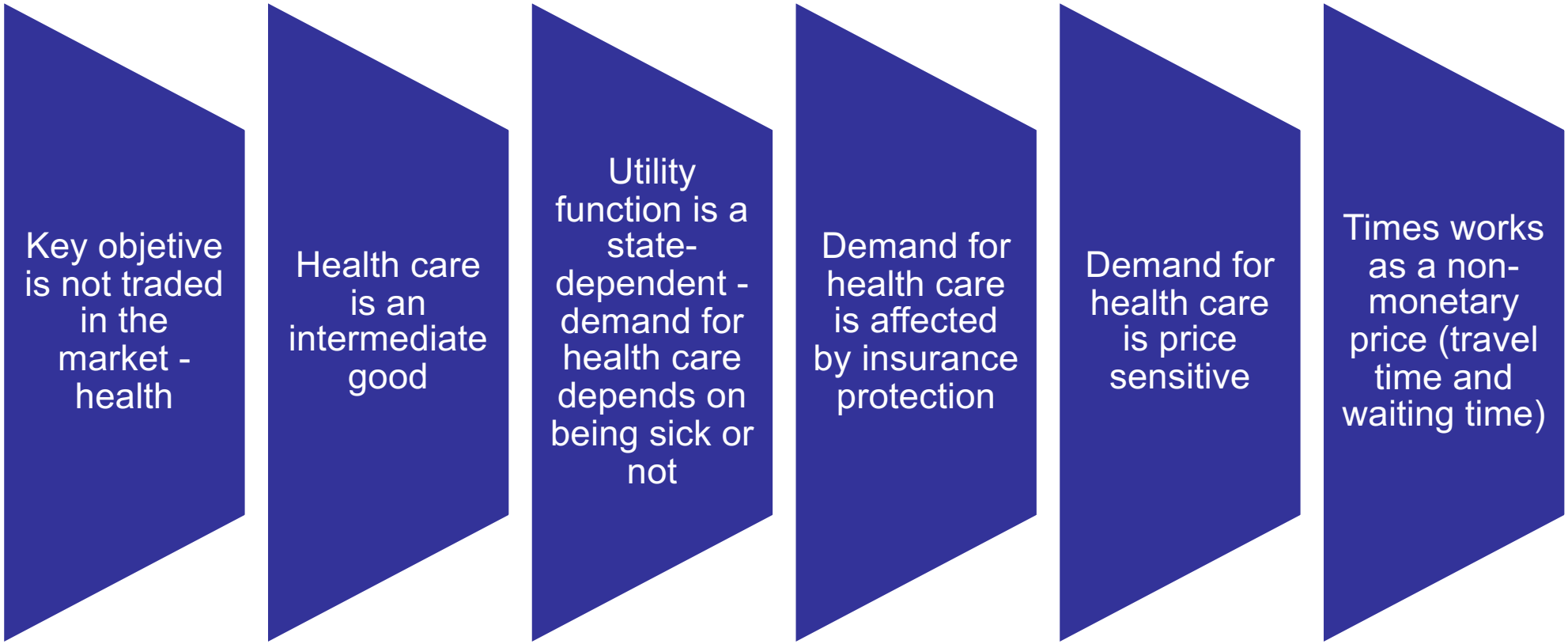
Price elasticities: evidence using Rand data

- Price elasticity estimates
  - All expenditures:
    - Rosett e Huang, 1973 hospital expenditure: -0.35 to 1.5
    - Manning et al, 1987, all expenditures: -0.17 to -0.22
  - Physician services:
    - Wedig, 1988, visits: -0.35 to -0.16, depending on health status
  - Hospital services:
    - Manning et al, 1987, hospital admissions: -0.14 to -0.17

Insurance elasticities

- Price elasticity estimates at the firm level (higher due demand switching)
  - McCarthy 1985 - consultations:
    - -3.07 a -3.26
  - Feldman Dowd, 1986, hospitals:
    - -0.74 a -0.80

Summarising

- Health is produced, not acquired
- Demand for health care is derived from demand for health
- Demand for health care is sensitive to several drivers
  - Distance (time of travel)
  - Waiting time
  - Prices (measured by different insurance protection)
  - Shifters of health production (age and education)

What changes on the demand side?